

Lecture Note 3: Exact Hat Algebra in a Multi-Sector Eaton–Kortum Model with Input–Output Linkages

In this lecture note we are adding input-output linkages to the J -sector model in Lecture Note 2. Firms will produce varieties using intermediate inputs sourced from their own sector and/or other sectors in addition to labor. These input-output linkages are what allows a shock in one sector (say, a tariff on steel) to affect costs, prices, and trade patterns in sectors that are not directly targeted by the shock (say, automobiles), purely through the input-output structure of production.

1. Motivation

Lecture Note 2 introduced $J > 1$ sectors but kept production a "pure value-added" activity: labor was the only input, and $c_i^j = w_i$ was the same across every sector within a country indexed by i . This environment highlights the role of sectoral comparative advantage and labor reallocation without the added complexity of intermediate inputs.

Real production is not like this. A car is made of steel, glass, electronics, and rubber, in addition to the labor of assembling them, and many of those inputs are themselves produced using inputs from yet other sectors. This lecture note relaxes the pure value-added assumption: firms in sector j combine labor with a bundle of intermediate inputs from sector $k \in \{1, \dots, J\}$, including $k = j$. This is the "cost propagates through supply chains" mechanism flagged but not modeled in Lecture Note 2.

The added realism comes at a cost. The presence of input-output linkages breaks some of the closed-form results (in particular, the welfare formula) that made the pure value-added model in Lecture Note 2 so tractable.

2. The Environment

The economy is populated by N countries and J sectors as in Lecture Note 2. The household's preferences are the same as before.

Preferences. The representative household in country n has Cobb–Douglas preferences across sectors,

$$U_n = \prod_{j=1}^J (C_n^j)^{\alpha_n^j}, \quad \sum_{j=1}^J \alpha_n^j = 1,$$

and CES love-of-variety preferences within each sector, $C_n^j = \left(\int_0^1 c_n^j(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}$. Recall from Appendix A in Lecture Note 1 that σ will not appear in anything that follows.

Technology. The firm producing variety ω in sector j in country i draws a productivity $z_i^j(\omega)$ from the Fréchet distribution $F_i^j(z) = \exp(-T_i^j z^{-\theta^j})$. In contrast to production in the first two notes, each firm now combines labor **and** intermediate inputs from every sector to produce a particular variety ω in sector j :

$$y_i^j(\omega) = z_i^j(\omega) \left[\ell_i^j(\omega) \right]^{\gamma_i^{j,0}} \prod_{k=1}^J \left[m_i^{j,k}(\omega) \right]^{\gamma_i^{j,k}},$$

where $m_i^{j,k}(\omega)$ is the quantity of the sector- k composite good C_i^k that firm ω uses as an intermediate input, and the exponents satisfy

$$\gamma_i^{j,0} + \sum_{k=1}^J \gamma_i^{j,k} = 1.$$

The production function has constant returns to scale with respect to labor and intermediate inputs jointly.

$\gamma_i^{j,0}$ is the **value-added share** of sector j in country i (the total wage bill as a fraction of gross output) and $\gamma_i^{j,k}$ is the share of costs paid for intermediate inputs used in sector j and sourced from sector k . Setting $\gamma_i^{j,0} = 1$ (and hence $\gamma_i^{j,k} = 0$ for every k) recovers the pure value-added model of Lecture Note 2.

Costs. Because the production function is Cobb-Douglas in labor and the J intermediate bundles, standard cost minimization delivers a unit cost of the form

$$c_i^j = B_i^j (w_i)^{\gamma_i^{j,0}} \prod_{k=1}^J (P_i^k)^{\gamma_i^{j,k}},$$

where P_i^k is country i 's price index for the sector k composite good C_i^k and B_i^j is a constant depending only on the cost shares $\{\gamma_i^{j,0}, \gamma_i^{j,1}, \dots, \gamma_i^{j,J}\}$. B_i^j plays the same "harmless constant" role that A played for the price index in Lecture Note 1.

This is the central new object relative to Lecture Note 2: the cost of production in sector j now depends on the price indices of every sector, not just the country-wide wage w_i . This is the formal channel through which a shock to sector k 's prices spreads to every sector j that uses k as an input.

Trade costs. International trade is subject to sector-origin-destination-specific iceberg costs $d_{ni}^j \geq 1$, as in Lecture Note 2.

3. Trade Shares, Price Indices, and the "hat" Algebra

The trade share and price index formulae are identical to the corresponding expressions in Lecture Note 2, except that c_i^j now reflects the richer cost function of Section 2 above rather than simply w_i :

$$\pi_{ni}^j = \frac{T_i^j \left(c_i^j d_{ni}^j \right)^{-\theta^j}}{\Phi_n^j}, \quad \Phi_n^j = \sum_{k=1}^N T_k^j \left(c_k^j d_{nk}^j \right)^{-\theta^j},$$

and

$$P_n^j = A^j \left(\Phi_n^j \right)^{-1/\theta^j}.$$

The hat-algebra derivation for $\hat{\pi}_{ni}^j$ and $\hat{P}_n^j$ goes through exactly as in Lecture Notes 1 and 2:

$$\hat{\pi}_{ni}^j = \frac{\hat{T}_i^j \left(\hat{c}_i^j \hat{d}_{ni}^j \right)^{-\theta^j}}{\sum_{k=1}^N \pi_{nk}^j \hat{T}_k^j \left(\hat{c}_k^j \hat{d}_{nk}^j \right)^{-\theta^j}}, \quad \hat{P}_n^j = \left[\sum_{k=1}^N \pi_{nk}^j \hat{T}_k^j \left(\hat{c}_k^j \hat{d}_{nk}^j \right)^{-\theta^j} \right]^{-1/\theta^j}.$$

What is new is the expression for $\hat{c}_i^j$ itself. Taking hats of the cost function in Section 2 it is immediately clear that the constant B_i^j drops out, in the same way that A^j drops out $\hat{P}_n^j$:

$$\hat{c}_i^j = (\hat{w}_i)^{\gamma_i^{j,0}} \prod_{k=1}^J \left(\hat{P}_i^k \right)^{\gamma_i^{j,k}}.$$

This is the key linkage equation. It says that the change in sector j 's cost in country i is a Cobb-Douglas combination of the wage change and the price changes of every sector's output in country i . Since $\hat{P}_i^k$ is itself determined by cost changes $\hat{c}_i^k$ in every sector, the system is fully simultaneous across sectors within a country: one cannot solve for one sector's prices while holding the others fixed, even for a shock that directly hits only one sector.

4. Closing the Model (General Equilibrium)

Sectoral expenditures. Spending on sector j output in country n now has two sources: final consumption, and demand from every sector k using sector j as an intermediate input:

$$X_n^j = \alpha_n^j I_n + \sum_{k=1}^J \gamma_n^{k,j} R_n^k, \quad I_n \equiv w_n L_n + D_n.$$

As in Lecture Note 2, take $D_n \equiv 0$ for now.

Sectoral revenues. Revenues in sector j in country i are given by:

$$R_i^j = \sum_{n=1}^N \pi_{ni}^j X_n^j.$$

Labor market clearing. In contrast to Lecture Note 2, not all of a sector's revenue is a payment to labor, only the value-added share $\gamma_i^{j,0}$ is. The rest is paid out to (domestic and foreign) suppliers of intermediate inputs. Country i 's wage bill is therefore:

$$w_i L_i = \sum_{j=1}^J \gamma_i^{j,0} R_i^j.$$

General equilibrium in changes. Writing the counterfactual system in the same hats-and-primed notation as before:

$$R_i^{j'} = \sum_{n=1}^N \pi_{ni}^j \hat{\pi}_{ni}^j X_n^{j'}, \quad X_n^{j'} = \alpha_n^j (\hat{w}_n w_n L_n + D_n) + \sum_{k=1}^J \gamma_n^{k,j} R_n^{k'}, \quad \hat{w}_i w_i L_i = \sum_{j=1}^J \gamma_i^{j,0} R_i^{j'}.$$

Solving the system. The solution algorithm has an extra layer relative to Lecture Note 2, precisely because of the point made at the end of Section 3: for a *given* guess of wage changes $\{\hat{w}_i\}$, the price changes $\{\hat{P}_i^j\}_{j=1}^J$ within each country are themselves the solution to a simultaneous system across sectors (since $\hat{c}_i^j$ depends on every $\hat{P}_i^k$, which depends on every $\hat{c}_i^k$, and so on). A typical algorithm therefore nests two loops:

1. **Outer loop.** Guess a vector of wage changes $\{\hat{w}_i^{(0)}\}$.
2. **Inner loop.** Given the wage guess, solve the linked system of trade share and price index equations across *all* J sectors simultaneously (e.g., by iterating on $\{\hat{c}_i^j\} \rightarrow \{\hat{\pi}_{ni}^j\} \rightarrow \{\hat{P}_i^j\} \rightarrow \{\hat{c}_i^j\}$ until this

inner system converges).

3. Use the converged $\{\hat{\pi}_{ni}^j\}$ to compute $\{X_n^{j'}\}$ and $\{R_i^{j'}\}$, and check labor-market clearing.
4. Update the wage guess $\{\hat{w}_i^{(1)}\}$ and repeat the outer loop until it, too, converges.

The data required to calibrate this model must be richer than those laid out in Lecture Note 2: sector-level baseline trade shares π_{ni}^j , consumption shares α_n^j , sectoral trade elasticities θ^j , and the full input-output matrix $\{\gamma_i^{j,k}\}_{j,k=1}^J$ for every country i (in practice, taken from national input-output tables).

5. Welfare

It is tempting to guess that the welfare formula simply carries over from Lecture Note 2. It does not, and working out exactly why is instructive. Before doing so, it is worth revisiting the formula in Lecture Note 2 first.

A correction to Lecture Note 2. Re-deriving that formula carefully (using the boxed trade-share equation together with $d_{nn}^j = 1$, so that $\hat{\pi}_{nn}^j = \hat{T}_n^j (\hat{w}_n)^{-\theta^j} / \left[\sum_k \pi_{nk}^j \hat{T}_k^j (\hat{w}_k d_{nk}^j)^{-\theta^j} \right]$, and noting that the bracketed term is exactly $(\hat{P}_n^j)^{-\theta^j}$) gives $\hat{P}_n^j = \hat{w}_n (\hat{T}_n^j)^{-1/\theta^j} (\hat{\pi}_{nn}^j)^{1/\theta^j}$. Assuming no direct shock to country n 's own sectoral productivity ($\hat{T}_n^j = 1$), aggregating across sectors with the Cobb–Douglas weights, and using $\hat{W}_n = \hat{w}_n / \hat{P}_n$, the $\hat{w}_n$ terms **cancel completely**, leaving:

$$\hat{W}_n = \prod_{j=1}^J (\hat{\pi}_{nn}^j)^{-\alpha_n^j / \theta^j}.$$

This differs from the formula stated in Lecture Note 2 (which has an extra factor of $\hat{w}_n$ and the opposite sign on the exponent). The corrected version above is the direct multi-sector analogue of Lecture Note 1's $\hat{W}_n = (\hat{\pi}_{nn})^{-1/\theta}$: it is a weighted geometric average of sector-level home-share formulas, with weights α_n^j / θ^j , and — just as in the one-sector case — the wage change drops out entirely. The next edition of these lecture notes will correct this mistake in the second note.

Why input-output linkages break this. The cancellation above relied on $\hat{c}_n^j = \hat{w}_n$, which is the simplification that made Lecture Note 2's cost function trivial. With input-output linkages, $\hat{c}_n^j = (\hat{w}_n)^{\gamma_n^{j,0}} \prod_k (\hat{P}_n^k)^{\gamma_n^{j,k}}$ instead. Repeating the same steps (using $\hat{\pi}_{nn}^j$ and $d_{nn}^j = 1$) gives, for each sector j :

$$\hat{P}_n^j = \hat{c}_n^j (\hat{T}_n^j)^{-1/\theta^j} (\hat{\pi}_{nn}^j)^{1/\theta^j} = (\hat{w}_n)^{\gamma_n^{j,0}} \prod_{k=1}^J (\hat{P}_n^k)^{\gamma_n^{j,k}} \cdot (\hat{T}_n^j)^{-1/\theta^j} (\hat{\pi}_{nn}^j)^{1/\theta^j}.$$

This is no longer a formula that expresses $\hat{P}_n^j$ directly in terms of observables: the unknowns $\{\hat{P}_n^k\}_{k=1}^J$ appear on both sides of the equation, coupled together through the country's own input-output matrix $\{\gamma_n^{j,k}\}$. Taking logs, this is a *linear system* in $\{\log \hat{P}_n^j\}_{j=1}^J$:

$$\log \hat{P}_n^j = \gamma_n^{j,0} \log \hat{w}_n + \sum_{k=1}^J \gamma_n^{j,k} \log \hat{P}_n^k + \frac{1}{\theta^j} \log \hat{\pi}_{nn}^j - \frac{1}{\theta^j} \log \hat{T}_n^j,$$

which can be solved in closed form via a Leontief inverse of the country's input-output matrix, but this is no longer a simple product of home trade share. Rather, the exponents that end up multiplying each $\log \hat{\pi}_{nn}^j$ term after solving the linear system depend on the entire input-output structure $\{\gamma_n^{j,k}\}_{j,k}$, not just on α_n^j / θ^j as before.

Once intermediate inputs are added, there is, in general, no shortcut around solving the full general equilibrium system of Section 4 to compute welfare. The sector-specific home trade shares are still part of the calculation, but they enter the welfare calculation in a way that takes the input-output structure into account, not just the final-consumption weights α_n^j .

Under certain assumptions — most notably, a common trade elasticity $\theta^j = \theta$ across all sectors — Arkolakis, Costinot, and Rodríguez-Clare (2012) show that a clean sufficient-statistic result can be recovered, using an appropriately defined domestic share of gross expenditure (final consumption plus intermediate use) rather than final consumption alone. Deriving that special case is a good extension exercise, but it relies on the common- θ assumption in a way the general case here does not.

6. What is the Value Added of Including Input-Output Linkages?

Relative to Lecture Note 2, input-output linkages add a channel through which sector-specific shocks propagate through other sectors within a country, even absent any wage response. In Lecture Note 2, a trade-cost shock confined to sector 1 could only affect sector 2's trade pattern through the single countrywide wage $\hat{w}_i$ (i.e., through the general equilibrium effect).

Here, it can also affect sector 2 directly through $\hat{P}_i^1$, whenever sector 2 uses sector 1 as an input ($\gamma_i^{2,1} > 0$) — this is true even in a hypothetical world with a fixed wage. This is precisely the "tariff on steel raises the cost of cars" mechanism motivated in Section 1, and it is also the reason the clean sufficient-statistic welfare result of the earlier lecture notes does not survive unmodified: once sectors are linked on the cost side, a full accounting of how much of every sector's cost increase is attributable to home-produced versus imported inputs is unavoidable.